

Experiment 6

Name: Urvi Shivalkar

PRN: B25ET1196

Class: SY-3(C)

Title: Implement stack using linked list

Code:

```
#include <stdio.h>
#include <stdlib.h>
struct Node {
    int data;
    struct Node* next;
};
struct Node* top = NULL;

void insertNode(int value) {
    struct Node* newNode = (struct Node*)malloc(sizeof(struct Node));
    if (newNode == NULL) {
        printf("Stack Overflow! Memory allocation failed.\n");
        return;
    }
    newNode->data = value;
    newNode->next = top;
    top = newNode;
    printf("%d inserted into the stack.\n", value);
}

void deleteNode() {
    if (top == NULL) {
        printf("Stack Underflow! The stack is empty.\n");
        return;
    }
    struct Node* temp = top;
    printf("%d deleted from the stack.\n", temp->data);
    top = top->next;
    free(temp);
}
```

```

void displayStack() {
    if (top == NULL) {
        printf("The stack is empty.\n");
        return;
    }
    struct Node* temp = top;
    printf("Stack elements (Top to Bottom):\n");
    while (temp != NULL) {
        printf("%d\n", temp->data);
        temp = temp->next;
    }
}

int main() {
    int choice, value;
    do {
        printf("\n--- Stack Operations Menu ---\n");
        printf("1. INSERT (Push)\n");
        printf("2. DELETE (Pop)\n");
        printf("3. DISPLAY\n");
        printf("4. Exit\n");
        printf("Enter your choice: ");
        scanf("%d", &choice);

        switch (choice) {
            case 1:
                printf("Enter value to insert: ");
                scanf("%d", &value);
                insertNode(value);
                break;
            case 2:
                deleteNode();
                break;
            case 3:
                displayStack();
                break;
            case 4:

```

```

        printf("Exiting program.\n");
        break;
    default:
        printf("Invalid choice! Please try again.\n");
    }
} while (choice != 4);

return 0;
}

```

Output:

```

--- Stack Operations Menu ---
1. INSERT (Push)
2. DELETE (Pop)
3. DISPLAY
4. Exit
Enter your choice: 1
Enter value to insert: 12
12 inserted into the stack.

```

```

--- Stack Operations Menu ---
1. INSERT (Push)
2. DELETE (Pop)
3. DISPLAY
4. Exit
Enter your choice: 1
Enter value to insert: 45
45 inserted into the stack.

```

```

--- Stack Operations Menu ---
1. INSERT (Push)
2. DELETE (Pop)
3. DISPLAY
4. Exit
Enter your choice: 1
Enter value to insert: 56
56 inserted into the stack.

```

```

--- Stack Operations Menu ---
1. INSERT (Push)
2. DELETE (Pop)
3. DISPLAY
4. Exit
Enter your choice: 1
Enter value to insert: 890
890 inserted into the stack.

```

```

--- Stack Operations Menu ---
1. INSERT (Push)
2. DELETE (Pop)
3. DISPLAY
4. Exit
Enter your choice: 3
Stack elements (Top to Bottom):
890
56
45
12

```

```

--- Stack Operations Menu ---
1. INSERT (Push)
2. DELETE (Pop)
3. DISPLAY
4. Exit
Enter your choice: 4
Exiting program.

```